Physiological
Measurement

TOPICAL REVIEW • OPEN ACCESS

Machine learning to detect, stage and classify diseases and their symptoms based on inertial sensor data: a mapping review

To cite this article: Daniele Bibbo *et al* 2023 *Physiol. Meas.* **44** 12TR01

View the article online for updates and enhancements.

You may also like

- Machine learning-based motor assessment of Parkinson's disease using postural sway, gait and lifestyle features on crowdsourced smartphone data
Hamza Abujrida, Emmanuel Agu and Kaveh Pahlavan

- Activity classification based on inertial and barometric pressure sensors at different anatomical locations
A Moncada-Torres, K Leuenberger, R Gonzenbach et al.

- Accelerometer techniques for capturing human movement validated against direct observation: a scoping review
Elyse Letts, Josephine S Jakubowski, Sara King-Dowling et al.

This content was downloaded from IP address 103.4.76.208 on 17/04/2026 at 17:41

IOP Publishing Physiol. Meas. 44 (2023) 12TR01 https://doi.org/10.1088/1361-6579/ad133b

Physiological Measurement

IPEM Institute of Physics and Engineering in Medicine

CrossMark

OPEN ACCESS

RECEIVED
19 June 2023

REVISED
8 November 2023

ACCEPTED FOR PUBLICATION
7 December 2023

PUBLISHED
26 December 2023

Original content from this work may be used under the terms of the Creative Commons Attribution 4.0 licence.

Any further distribution of this work must maintain attribution to the author(s) and the title of the work, journal citation and DOI.

TOPICAL REVIEW

Machine learning to detect, stage and classify diseases and their symptoms based on inertial sensor data: a mapping review

Daniele Bibbo[1,*], **Cristiano De Marchis**[2,*], **Maurizio Schmid**[1] and **Simone Ranaldi**[1]

[1] Department of Industrial, Electronic and Mechanical Engineering, Roma Tre University, Rome, Italy
[2] Department of Engineering, University of Messina, Messina, Italy
[*] These authors equally contributed to the article.

E-mail: cristiano.demarchis@unime.it

Keywords: clinical decision support system, inertial measurement unit, artificial intelligence, wearable technology

Abstract

This article presents a systematic review aimed at mapping the literature published in the last decade on the use of machine learning (ML) for clinical decision-making through wearable inertial sensors. The review aims to analyze the trends, perspectives, strengths, and limitations of current literature in integrating ML and inertial measurements for clinical applications. The review process involved defining four research questions and applying four relevance assessment indicators to filter the search results, providing insights into the pathologies studied, technologies and setups used, data processing schemes, ML techniques applied, and their clinical impact. When combined with ML techniques, inertial measurement units (IMUs) have primarily been utilized to detect and classify diseases and their associated motor symptoms. They have also been used to monitor changes in movement patterns associated with the presence, severity, and progression of pathology across a diverse range of clinical conditions. ML models trained with IMU data have shown potential in improving patient care by objectively classifying and predicting motor symptoms, often with a minimally encumbering setup. The findings contribute to understanding the current state of ML integration with wearable inertial sensors in clinical practice and identify future research directions. Despite the widespread adoption of these technologies and techniques in clinical applications, there is still a need to translate them into routine clinical practice. This underscores the importance of fostering a closer collaboration between technological experts and professionals in the medical field.

1. Introduction

A wide range of diseases across various specialisation areas can cause the emergence of motor symptoms. According to the International Classification of Diseases (ICD-10), many disorders affecting the nervous system are grouped under the term 'Movement disorders' (World Health Organization 1992), suggesting that information necessary for detecting pathologies in these areas can be directly inferred from motor manifestations. Modifications to movement-related functions include, but are not limited to, changes in joint mobility or muscle strength. These may arise as the byproduct of diseases affecting other systems (e.g. the nervous or the cardiovascular system) or syndromes associated with cognitive or behavioral disturbances. Consequently, motor symptoms serve as established indicators in a variety of diagnostic procedures across various specialties.

1.1. Use of technology for motor symptoms detection and classification

For over sixty years, technology has been an integral part of the motor symptoms determination process (Dierssen *et al* 1961). The new century has witnessed a significant increase in the availability and usage of devices able to collect such information, benefitting from trends in miniaturization, reduced energy consumption for

© 2023The Author(s). Published on behalf of Institute of Physics and Engineering in Medicine by IOP Publishing Ltd

data recording and processing, and advancements in wireless communications. These elements altogether have virtually eliminated all barriers to capturing data in natural environments (Vijayan *et al* 2021).

In this context, the use of inertial measurement units (IMUs) for clinical applications is a relatively recent development (Morris and Paradiso 2003). Thanks to miniaturisation and technological advancements, IMUs have become increasingly small and lightweight, making them suitable for integration into wearable devices in clinical settings: IMUs have been employed to detect and classify tremors in patients with neurological disorders that affect movement; they can also monitor changes in gait patterns over time, and this proves valuable in assessing the effectiveness of treatments for mobility impairments resulting from conditions like stroke, Parkinson's disease or spinal cord injury. In addition, IMUs can be worn with minimal encumbrance for extended periods, enabling long-term monitoring of motor symptoms and gait patterns, and this provides valuable insights into disease progression and response to treatment.

1.2. Plugging in machine learning

The advancement in computational power has facilitated the integration of processing functions at the system level, allowing the incorporation of machine learning into the process (Haick and Tang 2021). In particular, machine learning algorithms trained with IMUs data have shown promising capabilities in detecting and classifying specific motor symptoms associated with different diseases (Patel *et al* 2009). This technological approach holds the potential to enhance patient care by providing more accurate and objective assessments of motor symptoms, thereby enabling the tracking of disease progression and aiding treatment decisions (Lim *et al* 2022). In addition, the use of wearable IMUs for data collection enables continuous monitoring of patients, offering a more comprehensive representation of their symptoms over time.

While machine learning has the potential to greatly improve the detection of diseases and the classification or prediction of symptoms, there are also several limitations and challenges to consider (Kubota *et al* 2016). One key challenge is the need for large and diverse datasets to train and validate the machine learning algorithms. Inertial sensor data can be highly variable, and it can be difficult to collect enough representative data to accurately describe all possible variations that may be encountered in real-world scenarios. In addition, the accuracy of the machine learning algorithms can be affected by factors such as sensor placement, device performance, and the presence of noise or interference in the data. It is also important to consider ethical and privacy concerns when collecting and analyzing sensitive health data and to ensure that the technology is used in a responsible and transparent manner.

1.3. Aim of the review

This work aims at systematically reviewing the literature published since 2012 focusing on the utilization of machine learning to support clinical decision-making via the analysis of data coming from wearable inertial sensors, either alone or in combination with other sources. By quantitatively analyzing a set of selected papers through the lenses of four research questions, the review article aims to uncover the trends, perspectives, strengths and limitations within the extensive body of literature pertaining to the use of machine learning in clinical practice. Given the observed diversity of application fields and decision-making objectives of the involved studies, we decided to leave aside the analysis on the numerical performance of the different adopted solutions.

2. Methods

The analysis of the existing literature on the topic has been carried out by formulating four key *research questions* that define the main instrumental and clinical aspects of the previous research. Additionally, four *pertinence indicators* were defined and employed to guide the inclusion and exclusion criteria.

2.1. Research questions

The following research questions aim to summarize the current status of the research advancements in integrating machine learning and inertial measurements for clinical applications.

2.1.1. RQ1—which pathologies, which movements?

This research question focuses on characterizing the pathologies typically studied with the use of IMUs. Moreover, together with the investigation of the clinical conditions, it explores the movements under analysis and the experimental protocols that are adopted.

2.1.2. RQ2—which technologies and setups, which protocols?
The second research question investigates the most commonly used technologies in the literature. It examines the types and number of IMUs used, as well as the possible inclusion of additional measurements coming from different sensors. The protocol for sensor placement is also analysed.

2.1.3. RQ3—which data processing schemes, and which features?
This third question delves into the data processing workflow applied to the raw inertial data, before feeding the relevant features into the ML model. This includes all the steps from data conditioning (e.g. denoising filters) to dimensionality reduction (e.g. feature extraction and feature selection).

2.1.4. RQ4—which ML techniques, and for which aims?
The last research question investigates the ML techniques employed for data analysis. It describes the techniques used and examines the prediction objectives, including the control on possible comparisons among different techniques or training strategies.

2.2. Searching strategy

2.2.1. Chosen databases
For the systematic review, three databases were queried, namely *Elsevier Scopus*, *Clarivate Web of Science* and *Pubmed*.

2.2.2. Boolean expression
The query string, in its *Pubmed* form, is reported in the following

```
(
 (
  (deep learning[Title/Abstract])
  OR
  (machine learning[Title/Abstract])
  OR
  (artificial intelligence[Title/Abstract])
  OR
  (ai[Title/Abstract])
  OR
  (ml[Title/Abstract])
 )
 AND
 (
  classif*[Title/Abstract]
  OR stag*[Title/Abstract]
  OR evaluat*[Title/Abstract]
  OR asses*[Title/Abstract]
 )
 AND
 (
  diseas*[Title/Abstract]
  OR patholog*[Title/Abstract]
  OR sympto*[Title/Abstract]
 )
)
AND
(
 (
  inertial[Title/Abstract]
  AND
  (
   measur*[Title/Abstract]
   OR
   (
    senso*[Title/Abstract]
    ORunit[Title/Abstract]
   )
  )
 )
 OR
```

(Continued.)

```
 imu[Title/Abstract]
 OR
 mimu[Title/Abstract]
 OR
 imus[Title/Abstract]
 OR
 accelerom*[Title/Abstract]
 OR
 gyroscop*[Title/Abstract]
 OR
 wearabl*[Title/Abstract]
)
```

This query was converted into the corresponding form for each of the other databases, and run on September 14, 2022.

The main feature of the querying strategies was that of including all the research papers that show a combination (either in the *abstract* or in the *title*) of the concepts of machine learning, pathology, classification and/or staging and IMUs at large. The three results lists were joined using the *doi* identifier as a key, thus selecting an article for the successive phase if it was present in at least one of the three research lists.

2.2.3. Inclusion and exclusion criteria

From the resulting overall list, additional filters were applied as in the following. Articles listed as review, book chapter or conference paper as document type were removed from subsequent analyses, to single out results that reported original research in a complete form. The identification of the search results to be excluded was made through the keys in the query results (e.g. the *document type* key in the *Scopus* results); additional checks on the remaining results were carried out manually to remove any article that met the exclusion criteria, and any duplicate.

2.3. Pertinence assessment indicators

To further refine the search results, a set of four pertinence assessment indicators (PAI) was defined. While these indicators do not directly reflect the quality evaluation of a research, they were introduced to quantify the relevance of each article within the scope of the review. Each article was assigned an integer value for each PAI, resulting in an ordered set of pertinence classes aligned with the four research questions described in the previous section. All the included papers were evaluated in terms of these indicators by at least two authors of this review, according to the rules given in the following. In cases where there was a non-univocal result from one particular indicator, the evaluation was discussed among all the authors.

PAI_1—dataset characteristics, based on the sample size, in terms of number of patients involved in the study, and presence of an adequate control group. As this review focuses on clinical applications, the absence of a sample of patients in the study directly determined the exclusion from the successive analysis. Levels were thus defined as follows:

0—absence of any patient group.

1—small number of patients ($N < 10$) and no control group.

2—small number of patients ($N < 10$) with a control group.

3—substantial group of patients ($10 \leqslant N < 30$) and no control group.

4—any of the following:

- Substantial group of patients ($10 \leqslant N < 30$) with a control group
- High number of patients ($N \geqslant 30$).

PAI_2—clinical impact, based on the clinical relevance and impact of the study. If the technique for classification was not instrumental for the diagnosis staging or classification of the pathology, the study was excluded from further analysis. Levels were defined as follows:

0—no classification or detection of clinical interest.

1—indirect use of the data/features to capture elements of low clinical relevance

2—indirect use of the data/features to detect or classify pathologies.

3—direct use of the data for the classification or detection of pathologies, without connection to clinical scales.

4—direct use of the data for the classification or detection of pathologies, with connection to clinical scales.

PAI_3—use and readiness level of IMUs, based on the role that technology plays in the study, and the readiness level of the technology that was used for it. This element was considered as an indirect indicator of the study replicability. If no inertial technology was used in the study, this was excluded from successive analysis. Levels were defined as follows:

0—absence of any IMU sensor/technology.

1—presence of at least one IMU sensor, without a thorough description of the protocol.

2—presence of at least one IMU sensor with a low technology readiness level (TRL).

3—presence of at least one IMU sensor with a high TRL (e.g. a commercial sensor), used in conjunction with other sensors.

4—presence of at least one IMU sensor with a high TRL, without the use of additional sensors.

PAI_4—impact and role of ML, based on the importance of the Machine Learning techniques on the final results. If no ML was used in the analysis pipeline, this caused exclusion from further evaluation. Levels were defined as follows:

0—absence of any ML technique.

1—single ML technique that is only partly related to the results (e.g. used only for feature selection/extraction).

2—single ML technique with a direct effect on the clinically relevant results.

3—comparison or composition of different ML techniques for predicting clinically relevant information.

The pertinence indicators for each paper were determined from the independent evaluations, or consensus values when needed. A study was then included in the further review process if both the following conditions were met:

- $PAI_i > 0$ for all $i = 1, 2, 3, 4$
- $\sum_{i=1}^{4} \mathrm{PAI}_i \geqslant 11.$

The manuscripts close to the decision boundary for the second condition (i.e. total sum of the PAI_i equal to 10 or 11) were discussed by all the authors for confirmation of their inclusion or exclusion in the analysis.

2.4. Manuscript analysis

The selected articles from phase I were then analysed in relation to the different research questions. Each manuscript was reviewed by at least one author, and the main takeaways were categorized into four classes, corresponding to the four research questions. Based on the inclusion criteria and the fulfillment of the first condition on the PAI_i, each manuscript provided at least one key finding for the RQs. Following these analyses, the RQ answers derived from the articles were discussed and summarized by all the authors.

3. Results

This section will be organized as follows: the first subsection will present the analysis of the studies resulting from the search query (I). It will include information on the total number of studies identified, the general characteristics of those meeting the inclusion criteria on the individual PAIs and those that surpassed the overall PAI threshold; section 3.2 will focus on the analysis of the individual PAIs for the included studies, highlighting

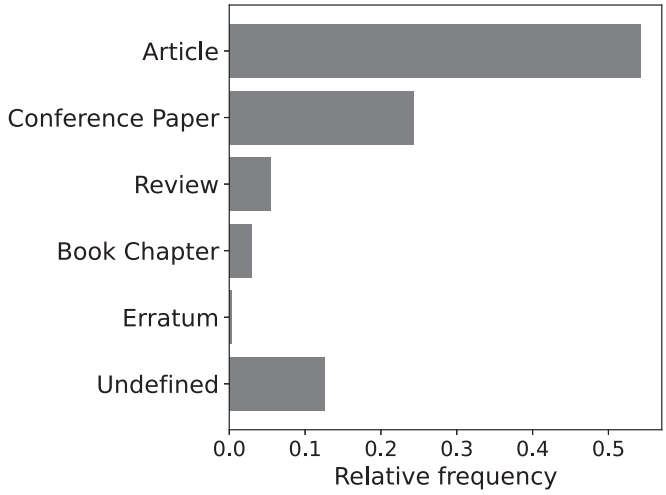

Figure 1. Article type of the query results.

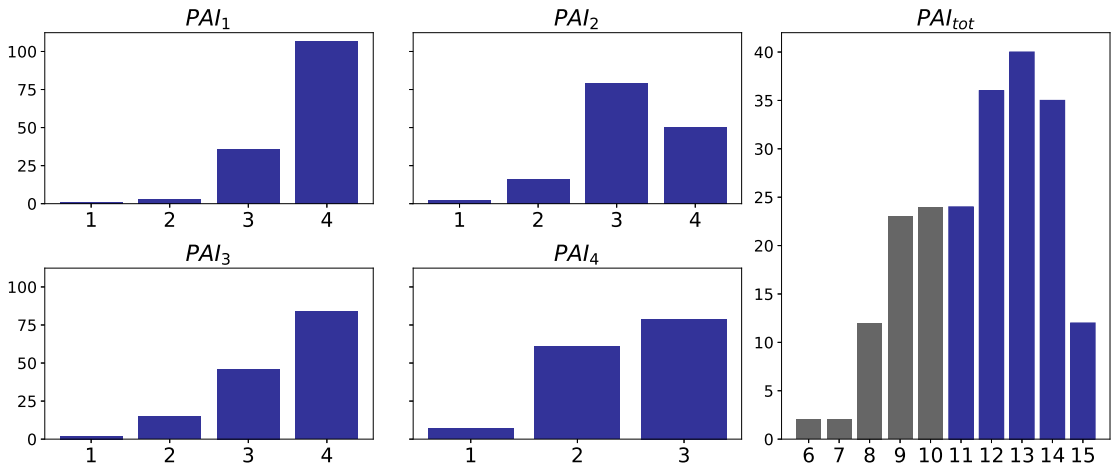

Figure 2. Left diagrams: histogram of PAI_i values for the above-threshold papers; right diagram: overall PAI score for papers with non-zero values, blue highlights papers with overall score above the threshold.

the distribution of the values assigned to each PAI; section 3.3 will provide an analysis of the studies that surpassed the threshold, with specific elements regarding each RQ.

3.1. Main characteristics of the phase I studies

A total number of 825 studies resulted from the query (see figure 1 for details on included studies). Out of these, 86 were excluded because they were not falling in the research article category, or not appearing in a Journal. Out of the remaining ones, 522 were also excluded because they had at least one zero for the PAI values. The total sum of PAI values was thus calculated on the remaining 217 research articles. Figure 2 shows the overall PAI value for this group. 146 articles yielded a value above the threshold, and underwent Phase II of the analysis.

3.2. Pertinence indicator results

3.2.1. On PAI_1

The vast majority (higher than 70%) of the above-threshold studies included a sample population size that was relevant for clinical classification purposes. A few studies were admitted to Phase II even if they obtained data from a reduced sample size (e.g. (Mostafa *et al* 2021)).

3.2.2. On PAI_2

Most studies that were admitted to Phase II made direct use of data for the classification, detection or staging of a disease or a symptom associated with it (aggregated percentage of 88% for $PAI_2 \geqslant 3$), with more than one third of them making direct reference to clinically agreed scales (e.g. Samà *et al* 2017).

3.2.3. On PAI_3

More than half of the above-threshold studies made use of commercially available IMUs alone, approximately one third used them in conjunction with other sensing technologies (Cole *et al* 2014, Johnson *et al* 2019, Chakraborty and Kishor 2022).

3.2.4. On PAI_4

Comparison of performance across different ML techniques was present in more than half of the studies that were analysed in Phase II. A few studies were above the threshold despite using ML technique only indirectly for classification or prediction (see e.g. Melin *et al* 2016, Hamy *et al* 2020).

3.3. Analysis of Phase II studies

Around three quarters of the papers were published in journals associated (also) with the Medicine subject area in Scopus SJR. Approximately half of the publications included Engineering and Computer Science, thus highlighting the inherent interdisciplinary nature of the topic. Other represented areas included Health Professions (approximately 22%) and Neuroscience (19%). In terms of specific Categories within each subject area, and disregarding miscellaneous ones (for each subject area, the category that groups together topics that cannot be assigned to a specific one), Electrical and Electronic Engineering and Health Informatics appeared as the most common ones, accounting for 30% and 25% of the papers, respectively. An interesting observation regarding the Medicine area is the absence of any specific clinical specialty category among the most prevalent ones. Only Clinical Neurology and Rehabilitation appeared in around 8% and 7% of the papers, respectively. Notably, the majority of papers published in journals with Medicine as one of the subject areas were categorized under the miscellaneous group.

3.3.1. RQ1—pathologies and movements

Regarding diseases, all the studies pertained to one of the following classes, defined according to the main affected system or organ, and grouped according to ICD-10:

- *Cardiovascular*diseases, which included congenital heart disease (Ganti *et al* 2022), atrial fibrillation (Mehrang *et al* 2020), myocardial infarction (Koivisto *et al* 2022) and congestive heart failure (Luštrek *et al* 2021). These are all included in Chapter IX of ICD-10.
- *Metabolic* diseases under study included Type I (Aljihmani *et al* 2022) and Type II (Lam *et al* 2021) Diabetes (ICD-10 Chapter IV).
- *Renal*conditions included studies targeting the chronic kidney disease (Bowlby *et al* 2016, West *et al* 2017, Glavinovic *et al* 2018).
- *Neurological*pathologies and conditions (Chapter VI of ICD-10) included Alzheimer's Disease (Corbi and Burgos 2021), Amyotrophic Lateral Sclerosis (Vieira *et al* 2022), cerebellar ataxia (Ngo *et al* 2021, Oubre *et al* 2021), Huntington's disease (Gordon *et al* 2019, Scheid *et al* 2022), Epilepsy (Zia *et al* 2021), Dementia, Essential tremor (Ma *et al* 2022a, 2022b) , Dystrophy (Maleki *et al* 2022), Mild Cognitive Impairment (Shahzad *et al* 2022), Multiple Sclerosis (Ibrahim *et al* 2020, Hossen *et al* 2022), and Parkinson's Disease (see e.g. (Caramia *et al* 2018, Cavallo *et al* 2019)).
- *Orthopedic* diseases (ICD-10 Chapter XIII) included Osteoarthritis (Moon *et al* 2020, Sun *et al* 2021), Rheumatoid Arthritis and Hip Arthroplasty (Dindorf *et al* 2020), among others (Ko *et al* 2021).
- *Psychiatric* disorders (ICD-10 Chapter V) included depression (Wahle *et al* 2016, Narziev *et al* 2020, Pedrelli *et al* 2020, Xu *et al* 2021), Post Traumatic Stress Disorder (Cakmak *et al* 2021), Children Internalizing Disorder (McGinnis *et al* 2019), Hyperactivity (Lindhiem *et al* 2022), and Anxiety (Jacobson *et al* 2020).
- *Respiratory* diseases (ICD-10 Chapter X) included Chronic Obstructive Pulmonary Disease (Gorzelniak *et al* 2014, Moy *et al* 2014), Idiopathic Pulmonary Fibrosis (Bahmer *et al* 2017) and Sleep Apnea Syndrome (Bricout *et al* 2021), or related disorders (Papini *et al* 2020). For this category, some studies included people with Covid-19 (Hassantabar *et al* 2021, Mason *et al* 2022, Skibinska and Burget 2022).

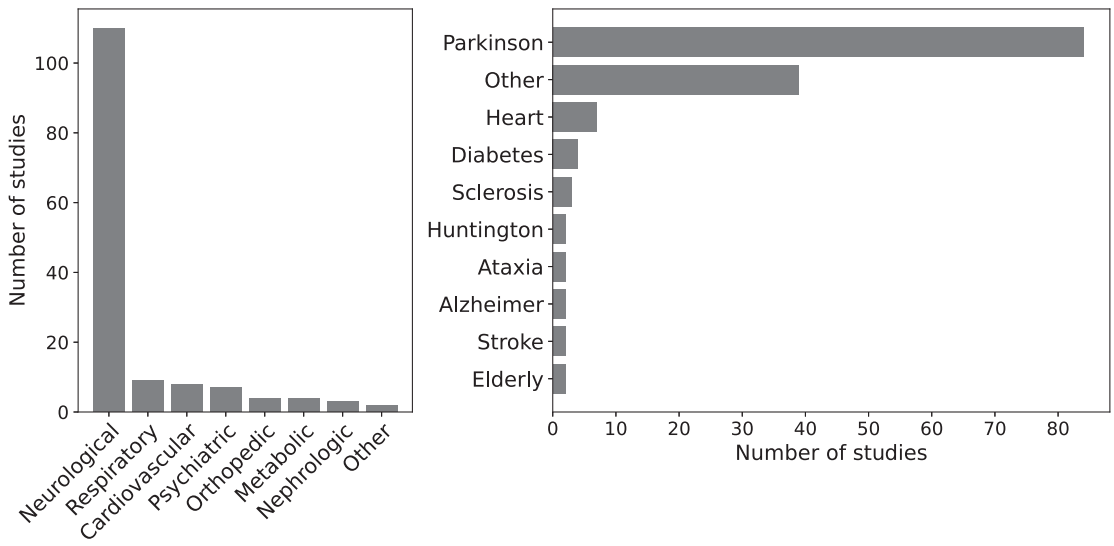

Figure 3. Disease category and specific pathology nature break-up of the analysed studies.

Among the identified categories, more than two thirds of the studies targeted neurological conditions, and more than half of them were related to Parkinson's Disease. Figure 3 includes the distribution of the studies across the categories, and a detailed view on specific clinical conditions studied.

For what concerns the analyzed task, all the studies were conducted under experimental conditions that pertained to one of the following categories:

- *Gait*: any form of locomotion, including straight overground walking (see e.g. Aich *et al* 2018, Cannière *et al* 2020, Trabassi *et al* 2022), curved walking, turning (Rehman *et al* 2020, Pardoel *et al* 2021) and treadmill walking (Romijnders *et al* 2022);
- *Upper Limb (UL)*: any form of non-functional exercise that specifically targeted movements of the upper limb or hands (see e.g. Huo *et al* 2020, Kim *et al* 2020, Park *et al* 2021);
- *Lower Limb (LL)*: any form of non-functional exercise that specifically targeted movements of the feet (Rovini *et al* 2018) or lower limbs in general (Rovini *et al* 2020);
- *Posture*: any form of task that requires the maintenance of a specific posture of the whole body (e.g. standing balance, (Memar *et al* 2017, Sotirakis *et al* 2022)) or of a body part (e.g. keep the arms extended in front of the body, (Xing *et al* 2022));
- *Rest*: any condition in which the experiment is carried out with the subject in resting state condition: e.g. rest tremor evaluation (de Araújo *et al* 2020), seated positions, lying supine positions, or sleep analysis;
- *Sit-To-Stand (STS)*: any experimental condition where the task or part of the task involves standing up or sitting down on a chair (see e.g. Hubble *et al* 2016, Tabatabaei *et al* 2020, Borzi *et al* 2022);
- *Activities of Daily Living (ADL)*: any experimental condition that targets functional activities or daily living ones (Lonini *et al* 2018, Li *et al* 2019), conducted in either controlled (Talitckii *et al* 2021) or uncontrolled conditions (Cook *et al* 2015, San-Segundo *et al* 2020, Wasselius *et al* 2021).

More than one third of the studied involved gait, and approximately the same share appeared for the activities of daily living. Less represented were the non-functional exercises.

3.3.2. RQ2—technologies, setups and protocols

More than half of the studies included only one IMU sensor, and just around 25% of the studies used more than 2 IMUs. Regarding the TRL, for those studies that disclosed specific characteristics of the used sensors, approximately three quarters made use of off-the-shelf products (Thomas *et al* 2018, Moon *et al* 2020, Teufl *et al* 2021), including those IMUs that were embedded into smartwatches, wristbands or smartphones (Sajal *et al* 2020). A few studies were made based on datasets already available (e.g. Palmerini *et al* 2017, Demrozi *et al* 2020, Ghosh and Banerjee 2021, Noor *et al* 2021, Park *et al* 2022). Regarding the chosen setups, the outlook was rather diversified, with around one third of the studies using IMUs placed only in the upper limb portion of the body

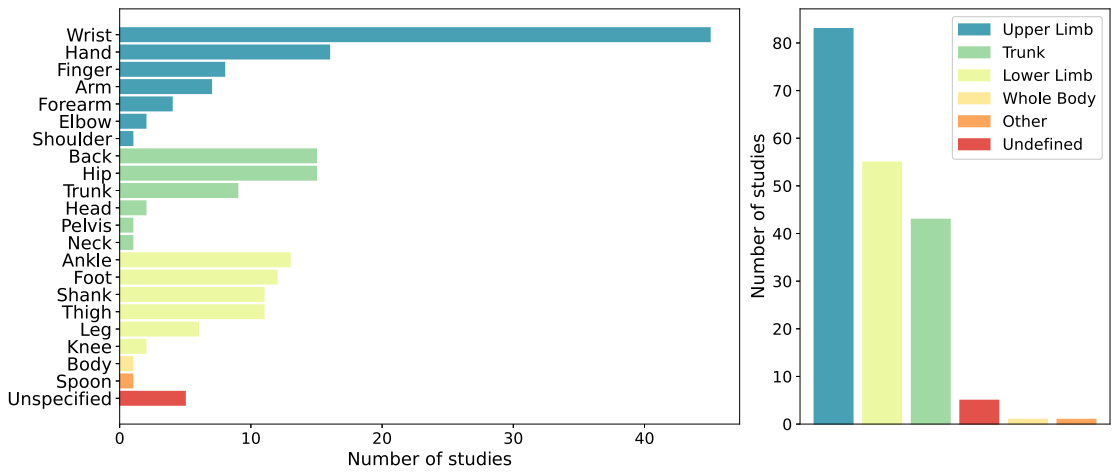

Figure 4. Placement locations for the inertial sensors.

(including wrists, e.g. in Wu *et al* 2021), and one fifth having the IMUs placed on the trunk only (Rodríguez-Martín *et al* 2015, Rehman *et al* 2019). Around 20% of the studies included multiple locations in different portions of the body (Hssayeni *et al* 2019, Butt *et al* 2020, Kuruvithadam *et al* 2021). Specific characteristics of the IMU placement are given in figure 4.

3.3.3. RQ3—data processing and features

The selected studies mostly relied on various sets of features to compactly describe the data gathered from the IMU sensors; a minor part of the studies skipped this step (Mannini *et al* 2016, Camps *et al* 2018), as this was implicitly performed by the machine learning part (such as in the case of deep ML models). When the feature extraction part was instead present, the extracted features were fed to the ML models described in section 3.3.4. Before feature extraction, the vast majority (higher than 65%) of the studies that provided details on the processing pipeline applied a standard denoising scheme for preserving relevant information, which was mostly based on low pass filtering with task-dependent cut-off frequencies, adjusted on the frequency content of the recorded signals. Feature sets were then extracted from fixed-length segments of the signal through a dedicated windowing procedure in around three quarters of the studies who provided details on this. A minor, yet not irrelevant, portion of the pipelines (around 20%) provided details on dimensionality reduction criteria for feature selection (see e.g. Jeon *et al* 2017, Shawen *et al* 2020). Regardless of the standard pre-processing steps, the extracted feature sets were here assigned to one of the following categories:

- *Aspecific*: general features that statistically describe the signal either in the time domain or in transformed domains, such as amplitude or spectral distribution parameters, without a direct specific link with the characteristics of the studied pathology, task, or type of signal.
- *Handcrafted*: features that are defined to capture fundamental or relevant aspects of the targeted pathology, of the used motor task, or of the recorded signal (e.g. power in the tremor frequency bands for PD, or spatiotemporal gait parameters for walking tasks).
- *Raw*: the denoised raw version of the signal is used as input to the deep ML model.

The selected studies used in comparable proportion sets of either aspecific (43%) or handcrafted (53%) features, with few (6%) using a mix of them (such as in Madrid-Navarro *et al* 2018, Rehman *et al* 2020, Donisi *et al* 2021). Only 10% of the studies relied on the raw version of the gathered signal, mainly for subsequent use in deep learning models (see e.g. Park *et al* 2017, Papadopoulos *et al* 2020, Wang *et al* 2022), where the feature extraction and selection is automatically assigned to dedicated layers of the deep neural network (Kim *et al* 2018). Concerning the specific nature of the extracted features, a focus on the most common scenario found in the literature (i.e. the classification of Parkinson's disease or its symptoms) shows that there are no clear trends on the nature of the information that is captured from IMU data, with a trend towards the extraction of a large set of heterogeneous features. In particular, more than 60% of the studies extracted amplitude-based features (such as ranges, averages, parameters of variability, peak values) in the time domain or in the frequency domain (mean frequency, energy in specific spectral ranges, ...), and more than 20% included also indicators of complex

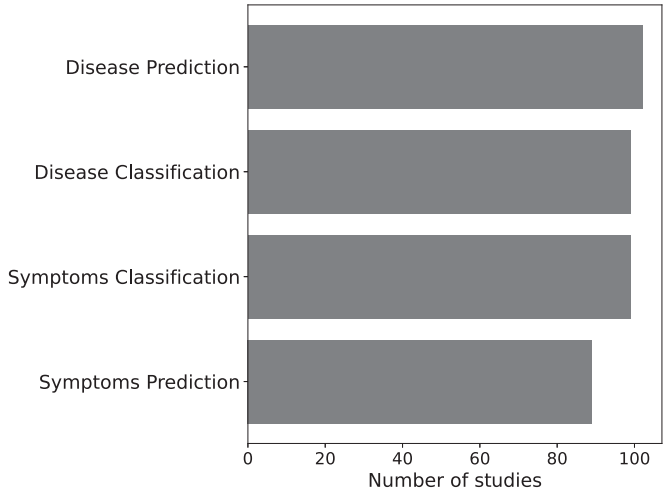

Figure 5. Aim of the ML procedure.

dynamics (such as sample entropy or approximate entropy). Just a reduced share (less than 25% of the studies) limited to one specific domain.

3.3.4. RQ4—machine learning techniques and classification outcomes

- Starting from the features set described in section 3.3.3, the aim of the ML techniques adopted within the selected studies was either a prediction or a classification problem (see figure 5). The final aim of the ML models with reference to the disease described in section 3.3.1 was either to predict or classify the samples based on specific symptoms associated with a disease (for instance, freezing of gait in Parkinson's Disease in Xia *et al* (2018) and Rodríguez-Martín *et al* (2017)) or the general presence of a pathological condition. All those studies adopting a regression model aimed at predicting symptoms, according to self-defined scores or accepted clinical scales. Those studies using ML for solving a classification problem were either aimed at classifying the symptoms (Darnall *et al* 2012, Ahlrichs *et al* 2015) or the presence of a pathology (Vos *et al* 2020, Yin *et al* 2021, Kovalenko *et al* 2022). In one interesting case, performance was evaluated across different environment conditions entailing the same tasks, i.e. in real world as compared to lab (Rehman *et al* 2022). For what concerns the used ML techniques, all the studies adopted at least one method that pertained to one of the following categories:

- *ANN* : all kinds of non-deep artificial neural networks (Oung *et al* 2017, Hssayeni *et al* 2019), including ANFIS.
- *DNN* : all kinds of deep learning (e.g. Bidabadi *et al* 2019, Hssayeni *et al* 2021), and long short-term memory networks (Sigcha *et al* 2020).
- *DT*: all kinds of Decision tree learning (e.g. in Khodakarami *et al* 2019, Talitckii *et al* 2022), with the inclusion of the regression trees (CART).
- *k-NN* : k-nearest neighbour models (Borzi *et al* 2021, Dai *et al* 2021, Mesin *et al* 2022);
- *LR*: Logistic, linear regression and all kinds of discriminant analyses (LR, LDA, DA) such as in (Kheirkhahan *et al* 2016).
- *RF*: all kinds of ensemble learning, including bagging, boosting, random forest (Kuhner *et al* 2020, Donisi *et al* 2021, Mirelman *et al* 2021).
- *NB*: Naive Bayes-based and other Bayesian classifying schemes (e.g. Cuzzolin *et al* 2017, Mileti *et al* 2018), including those based on temporal prediction such as HMM;
- *SVM* : vector machines, such as relevance vector machines, support vector machines (Bernad-Elazari *et al* 2016, Oliveira *et al* 2018, Dvorani *et al* 2021).
- *ZZ-Oth*: models not categorised elsewhere (Watts *et al* 2021), and included hard clustering and fuzzy clustering with supervision (Lonini *et al* 2018, Dvorani *et al* 2021).

Around 56% of the studies applied more than one ML technique (Aich *et al* 2020) and usually compared the results (Samà *et al* 2018, Donohue *et al* 2020), while the other studies relied on the use of a single ML technique.

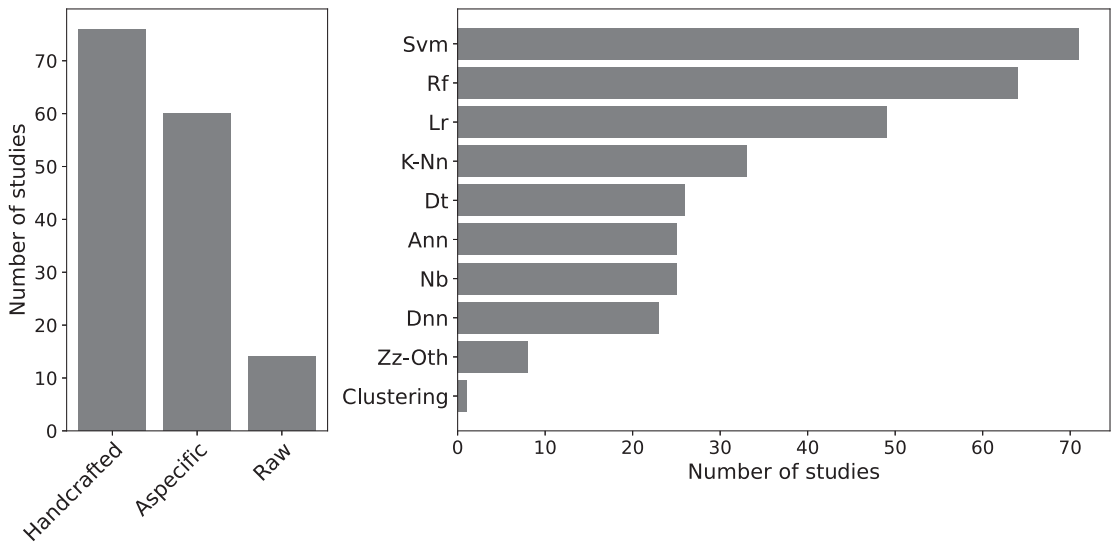

Figure 6. Feature extraction strategies and adopted ML algorithms.

The right panel of figure 6 shows the relative proportion of the adopted ML techniques. Vector machines and ensemble learning models resulted as the most used ML techniques.

Regarding the specific objective of the ML models, around 42% of the studies aimed at the classification of a disease (e.g. Costa *et al* 2016, Butt *et al* 2017, Williamson *et al* 2021), and 29% were designed to isolate the presence of a condition (such as a symptom) of clinical interest (for instance, Channa *et al* 2021, Sigcha *et al* 2021); approximately 35% of the studies used the ML models to predict scores or values of clinically relevant or agreed scales (e.g. Wan *et al* 2018, Borzi *et al* 2020, Ramesh and Bilal 2022).

4. General discussion

The majority of studies focusses on monitoring Parkinson's Disease, highlighting the potential of combining non-intrusive wearable technologies with machine learning for staging and symptom detection. This focus can be attributed to the well-defined motor symptoms associated with Parkinson's disease, making it a suitable target for analysis using wearable devices. However, there are relatively fewer studies that investigate other high-incidence diseases that affect the central or peripheral nervous system, such as Stroke, Alzheimer's Disease, or Muscular Dystrophies, despite the presence of motor symptoms in these conditions. The scarcity of studies on these diseases may be mainly attributed to the inherent variability of motor symptoms, of their clinical course and progression, potentially making the practical use of wearable technologies combined with ML for these specific conditions more challenging. Interestingly, some studies have identified psychiatric disorders as an area of interest. This suggests the potential utility of these technologies in capturing higher-level behavioral information beyond motor tasks. The inclusion of psychiatric disorders in the research focus underscores the broad application of wearable devices beyond traditional neurological conditions, and expanding their potential impact on understanding and managing a wider range of disorders.

Regarding the use of sensors, more than half of the studies included in the review used only one IMU. This finding suggests that a single-sensor setup generally provides sufficient information for obtaining satisfactory results when combined with the selected machine learning algorithms. This preference for a simple and practical setup is particularly relevant in clinical contexts and with patients, where the usability of the system plays a fundamental role. In terms of the technologies adopted and their readiness level, a significant proportion of the studies used off-the-shelf products that incorporated IMUs: this indicates that commercially available sensors, which can be easily integrated into wearable devices such as smartwatches, wristbands, or smartphones, are generally considered reliable and preferable due to their convenience and simplified setup. Although the placement of IMUs varied among the studies, there was a tendency towards simplified configurations. Approximately one-third of the studies focused on placing the IMU on the upper limb, while one-fifth placed it on the trunk only. This further supports the preference for simplified setups. In contrast, only a small share of the studies incorporated IMUs in multiple locations across different parts of the body, suggesting that more complex sensor configurations, including the use of multiple sensors or custom solutions, were not perceived as adding significant information to improve the implemented ML techniques. In summary, the overall findings concerning the technological aspects indicate a preference for a simplified setup, involving a small number of

commercially available sensors positioned in the upper part of the body. This approach is considered practical and effective for plugging in ML techniques.

For the processing pipeline, the methods for feature extraction showed common choices, both in the case of *aspecific* features, that have no direct physiological meaning, and in the case of *handcrafted* features, that in most of the cases represent standard parameters typically investigated in the clinical research, such as the spatiotemporal features of gait. Other processing steps in the feature extraction , when present, involve denoising filters and unsupervised feature selection and reductions. When advanced deep learning techniques were employed, feature extraction was embedded within the machine learning model itself. Considering the high number of data points that can be recorded by IMU sensors over an extended period of time, the engineering of ad-hoc features was found to be less critical. Only half of the analysed studies used engineered features, and this was often due to the integration of the ML model in a processing pipeline derived from the clinical research, which includes e.g. temporal segmentation and parameter calculation. Instead, researchers have leveraged the wealth of data captured by IMUs and incorporated established feature extraction methods, both generic and tailored to specific clinical parameters. This approach allows for efficient analysis and interpretation of the sensor data, with the integration of ML models playing a central role in exploiting the information contained within the features. Moreover, the current technologies allow for an easy processing of a large number of features; considering this, most of the choices that have been found in this review adopt a conservative strategy, in which the ML model works on several different features that describe most of the characteristics of the incoming signal, without the need for increasing *a priori* knowledge or computational complexity. It is predicted that this trend may be even more pronounced in the future, given the increased popularity of deep learning pipelines, where the process of feature extraction can be integrated into the ML model.

In terms of the problem ML seeks to solve, there is no clear tendency towards a specific use. Classification and regression techniques are applied equally to specific symptoms or general aspects of the pathology. This indicates that the current availability and variety of ML techniques allow for their flexible application across different clinical scenarios. ML methods can be adapted to specific clinical needs, highlighting the versatility of data recorded from IMUs, mainly consisting of features derived from acceleration and angular rate time series. Regarding the adopted ML technique, deep learning models were, at the time of the query execution, still infrequently used, suggesting that the trade-off between data structure (i.e time series from a few channels) and the computational burden of DL is unbalanced, even in large datasets. Most studies employ multiple ML approaches and compare their results. Among the individual ML techniques, there is a clear preference for ensemble learning models, despite the continued popularity of vector machines. This indicates that in the specific field analyzed in this review, there is no consensus or prior knowledge regarding which technique may outperform others. Even though beyond the aims of the present systematic review, it is worth noting that the variety of approaches in this field opens up opportunities for disease-specific meta-analysis studies focusing on the quantitative performance assessment of individual ML techniques.

To uncover potential relationships among the results obtained for the different research questions, an analysis using association rules was conducted, aiming to identify the most frequent co-occurrence of rules. As shown in figure 7, the use of machine learning for diagnosis and classification of nervous system diseases is often linked to the absence of additional sensors. This suggests that the information conveyed through intertial sensors alone is deemed sufficient for such classification or regression tasks. Secondly, when gait is chosen as the task for classification, researchers tend to utilize inertial sensors placed on the trunk without employing additional ones. This implies that trunk-based inertial sensors offer valuable information for gait analysis and classification, eliminating the need for other sensor modalities. Lastly, in terms of processing techniques, the feature extraction process typically occurs after temporal segmentation and/or filtering/denoising processes. This indicates that researchers often apply temporal segmentation or filtering/denoising techniques as preprocessing steps to refine the data before extracting relevant features for machine learning.

The studies included in the analysis were categorized based on the subject areas of the journals in which they were published. Using the Scimago Journal and Country Rank (SJR), a total of 15 subject areas were identified, as shown in figure 8. Surprisingly, despite the focus of the analyzed studies on the application of machine learning techniques for detecting or staging pathologies or symptoms, the majority of the articles were published in journals primarily associated with the scientific or technical field. In contrast, only a small percentage of the articles were published in journals related to the clinical field. This finding suggests that while machine learning techniques have garnered significant attention in clinical applications across various specialty areas, there are still barriers that need to be addressed to facilitate their widespread adoption in clinical practice.

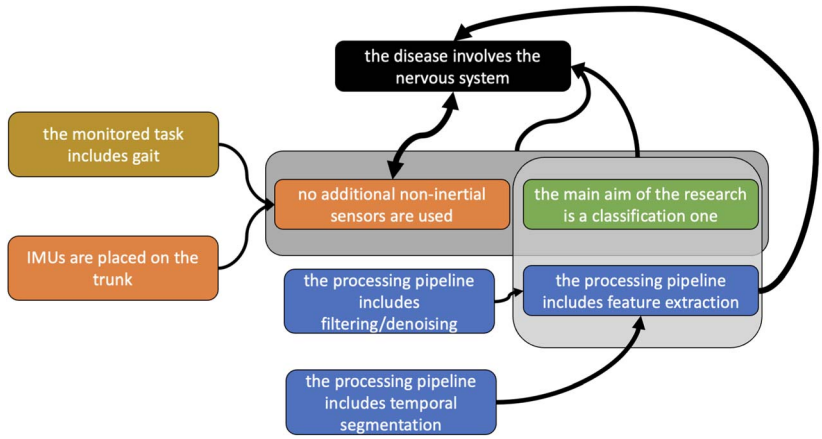

Figure 7. Main association rules obtained. Connection line width is proportional to rule confidence.

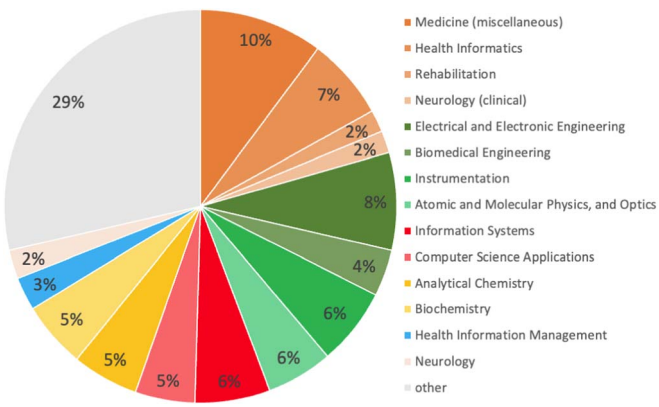

Figure 8. Shares of categories according to the SJR classification of the journals where the selected papers were published.

5. Conclusions

The integration of machine learning algorithms with lightweight inertial sensors has emerged as an important topic in the current scientific literature. The outcomes of this review work show that, for a variety of different pathologies, these technologies can have a wide impact on the detection and staging of the patients' conditions. It is however important to acknowledge that a solution that is universally accepted in the clinical practice is still to be fully developed. Nevertheless, the vast majority of the studies under analysis highlight that the integration of a wearable sensor, often directly included in consumer technological objects (e.g. smartphones, wristbands), with techniques that learn by experience can be a powerful tool in extracting information from the patient's behaviour during the activities of daily living, with little or no influence on the freedom of movement and quality of life. Moving forward, further interdisciplinary collaboration between researchers, clinicians, and technology developers is essential to address the remaining challenges and bridge the gap between scientific advancements and practical clinical applications.

Data availability statement

No new data were created or analysed in this study.

ORCID iDs

Daniele Bibbo https://orcid.org/0000-0003-1341-5427
Cristiano De Marchis https://orcid.org/0000-0003-1522-7454
Maurizio Schmid https://orcid.org/0000-0003-3380-6994

Simone Ranaldi https://orcid.org/0000-0002-7849-0893

References

Ahlrichs C *et al* 2015 Detecting freezing of gait with a tri-axial accelerometer in parkinson's disease patients *Med. Biol. Eng. Comput.* **54** 223–33

Aich S *et al* 2018 A validation study of freezing of gait (FoG) detection and machine-learning-based FoG prediction using estimated gait characteristics with a wearable accelerometer *Sensors* **18** 3287

Aich S *et al* 2020 Design of a machine learning-assisted wearable accelerometer-based automated system for studying the effect of dopaminergic medicine on gait characteristics of parkinson's patients *J. Healthcare Eng.* **2020** 1–11

Aljihmani L *et al* 2022 Hand tremor-based hypoglycemia detection and prediction in adolescents with type 1 diabetes *Biomed. Signal Process. Control* **78** 103869

Bahmer T *et al* 2017 Prognosis and longitudinal changes of physical activity in idiopathic pulmonary fibrosis *BMC Pulmonary Med.* **17** 104

Bernad-Elazari H *et al* 2016 Objective characterization of daily living transitions in patients with parkinson's disease using a single body-fixed sensor *J. Neurol.* **263** 1544–51

Bidabadi S S *et al* 2019 Classification of foot drop gait characteristic due to lumbar radiculopathy using machine learning algorithms *Gait Posture* **71** 234–40

Borzì L *et al* 2021 Prediction of freezing of gait in parkinson's disease using wearables and machine learning *Sensors* **21** 614

Borzì L *et al* 2022 Predicting axial impairment in Parkinson's disease through a single inertial sensor *Sensors* **22** 412

Borzi L *et al* 2020 Smartphone-based estimation of item 3.8 of the MDS-UPDRS-III for assessing leg agility in people with Parkinson's disease *IEEE Open J. Eng. Med. Biol.* **1** 140–7

Bowlby W *et al* 2016 Physical activity and metabolic health in chronic kidney disease: a cross-sectional study *BMC Nephrol.* **17** 187

Bricout A *et al* 2021 Accelerometry-derived respiratory index estimating apnea-hypopnea index for sleep apnea screening *Comput. Methods Programs Biomed.* **207** 106209

Butt A H *et al* 2017 Biomechanical parameter assessment for classification of parkinson's disease on clinical scale *Int. J. Distrib. Sens. Netw.* **13** 155014771770741

Butt A H *et al* 2020 Data-driven models for objective grading improvement of Parkinson's disease *Ann. Biomed. Eng.* **48** 2976–87

Cakmak A S *et al* 2021 Classification and prediction of post-trauma outcomes related to PTSD using circadian rhythm changes measured via wrist-worn research watch in a large longitudinal cohort *IEEE J. Biomed. Health Inform.* **25** 2866–76

Camps J *et al* 2018 Deep learning for freezing of gait detection in parkinson's disease patients in their homes using a waist-worn inertial measurement unit *Knowl. Syst.* **139** 119–31

Cannière H D *et al* 2020 Wearable monitoring and interpretable machine learning can objectively track progression in patients during cardiac rehabilitation *Sensors* **20** 3601

Caramia C *et al* 2018 IMU-based classification of Parkinson's disease from gait: a sensitivity analysis on sensor location and feature selection *IEEE J. Biomed. Health Inform.* **22** 1765–74

Cavallo F *et al* 2019 Upper limb motor pre-clinical assessment in parkinson's disease using machine learning *Parkinsonism Related Disorders* **63** 111–6

Chakraborty C and Kishor A 2022 Real-time cloud-based patient-centric monitoring using computational health systems *IEEE Trans. Comput. Soc. Syst.* **9** 1613–23

Channa A *et al* 2021 A-WEAR bracelet for detection of hand tremor and bradykinesia in Parkinson's patients *Sensors* **21** 981

Cole B T *et al* 2014 Dynamical learning and tracking of tremor and dyskinesia from wearable sensors *IEEE Trans. Neural Syst. Rehabil. Eng.* **22** 982–91

Cook D J, Schmitter-Edgecombe M and Dawadi P 2015 Analyzing activity behavior and movement in a naturalistic environment using smart home techniques *IEEE J. Biomed. Health Inform.* **19** 1882–92

Corbi A and Burgos D 2021 Connection between sleeping patterns and cognitive deterioration in women with alzheimer's disease *Sleep Breathing* **26** 361–71

Costa L *et al* 2016 Application of machine learning in postural control kinematics for the diagnosis of alzheimer's disease *Comput. Intell. Neurosci.* **2016** 1–15

Cuzzolin F *et al* 2017 Metric learning for parkinsonian identification from IMU gait measurements *Gait Posture* **54** 127–32

Dai H *et al* 2021 Validation of inertial sensing-based wearable device for tremor and bradykinesia quantification *IEEE J. Biomed. Health Inform.* **25** 997–1005

Darnall N *et al* 2012 Application of machine learning and numerical analysis to classify tremor in patients affected with essential tremor or Parkinson's disease *Gerontechnology* **10** 209–219

de Araújo A C A *et al* 2020 Hand resting tremor assessment of healthy and patients with parkinson's disease: An exploratory machine learning study *Front. Bioeng. Biotechnol.* **8** 778

Demrozi F *et al* 2020 Toward a wearable system for predicting freezing of gait in people affected by Parkinson's disease *IEEE J. Biomed. Health Inform.* **24** 2444–51

Dierssen G, Lorenc M and Spitaleri R M 1961 A new method for graphic study of human movements *Neurology* **11** 610–610

Dindorf C *et al* 2020 Interpretability of input representations for gait classification in patients after total hip arthroplasty *Sensors* **20** 4385

Donisi L *et al* 2021 Positive impact of short-term gait rehabilitation in parkinson patients: a combined approach based on statistics and machine learning *Mathematical Biosciences and Engineering* **18** 6995–7009

Donohue C *et al* 2020 A preliminary investigation of whether HRCA signals can differentiate between swallows from healthy people and swallows from people with neurodegenerative diseases *Dysphagia* **36** 635–43

Dvorani A *et al* 2021 Real-time detection of freezing motions in parkinson's patients for adaptive gait phase synchronous cueing *Front. Neurol.* **12** 720516

Ganti V G *et al* 2022 Wearable seismocardiography-based assessment of stroke volume in congenital heart disease *J. Am. Heart Assoc.* **11** e026067

Ghosh N and Banerjee I 2021 IoT-based freezing of gait detection using grey relational analysis *Internet Things* **13** 100068

Glavinovic T *et al* 2018 CKD and sedentary time: results from the canadian health measures survey *Am. J. Kidney Dis.* **72** 529–37

Gordon M F *et al* 2019 Quantification of motor function in huntington disease patients using wearable sensor devices *Digit. Biomarkers* **3** 103–15

Gorzelniak L *et al* 2014 Classification of exacerbation episodes in chronic obstructive pulmonary disease patients *Methods Inf. Med.* **53** 108–14
Haick H and Tang N 2021 Artificial intelligence in medical sensors for clinical decisions *ACS Nano* **15** 3557–67
Hamy V *et al* 2020 Developing smartphone-based objective assessments of physical function in rheumatoid arthritis patients: the PARADE study *Digit. Biomarkers* **4** 26–44
Hassantabar S *et al* 2021 CovidDeep: SARS-CoV-2/COVID-19 test based on wearable medical sensors and efficient neural networks *IEEE Trans. Consum. Electron.* **67** 244–56
Hossen A *et al* 2022 Machine learning aided classification of tremor in multiple sclerosis *eBioMedicine* **82** 104152
Hssayeni M D, Jimenez-Shahed J, Burack M A and Ghoraani B 2021 Ensemble deep model for continuous estimation of unified Parkinson'sdisease rating scale III *BioMed. Eng. OnLine* **20** 32
Hssayeni M D, Jimenez-Shahed J and Ghoraani B 2019 Hybrid feature extraction for detection of degree of motor fluctuation severity in Parkinson's disease patients *Entropy* **21** 137
Hubble R P *et al* 2016 Assessing stability in mild and moderate Parkinson's disease: Can clinical measures provide insight? *Gait Posture* **49** 7–13
Huo W *et al* 2020 A heterogeneous sensing suite for multisymptom quantification of Parkinson's disease *IEEE Trans. Neural Syst. Rehabil. Eng.* **28** 1397–406
Ibrahim A A *et al* 2020 Inertial sensor-based gait parameters reflect patient-reported fatigue in multiple sclerosis *J. Neuroeng. Rehabil.* **17** 165
Jacobson N C, Summers B and Wilhelm S 2020 Digital biomarkers of social anxiety severity: Digital phenotyping using passive smartphone sensors *J. Med. Internet Res.* **22** e16875
Jeon H *et al* 2017 Automatic classification of tremor severity in Parkinson's disease using a wearable device *Sensors* **17** 2067
Johnson A *et al* 2019 Use of mobile health apps and wearable technology to assess changes and predict pain during treatment of acute pain in sickle cell disease: feasibility study *MIR mHealth uHealth* **7** e13671
Kheirkhahan M *et al* 2016 Actigraphy features for predicting mobility disability in older adults *Physiol. Meas.* **37** 1813–33
Khodakarami H *et al* 2019 Prediction of the levodopa challenge test in Parkinson's disease using data from a wrist-worn sensor *Sensors* **19** 5153
Kim H B *et al* 2018 Wrist sensor-based tremor severity quantification in Parkinson's disease using convolutional neural network *Comput. Biol. Med.* **95** 140–6
Kim J Y *et al* 2020 Analysis of machine learning-based assessment for elbow spasticity using inertial sensors *Sensors* **20** 1622
Ko J B *et al* 2021 Predicting sarcopenia of female elderly from physical activity performance measurement using machine learning classifiers *Clin. Interventions Aging* **16** 1723–33
Koivisto T *et al* 2022 Mechanocardiography in the detection of acute ST elevation myocardial infarction: the MECHANO-STEMI study *Sensors* **22** 4384
Kovalenko E *et al* 2022 Detecting the Parkinson's disease through the simultaneous analysis of data from wearable sensors and video *IEEE Sensors J.* **22** 16430–9
Kubota K J, Chen J A and Little M A 2016 Machine learning for large-scale wearable sensor data in Parkinson's disease: Concepts, promises, pitfalls, and futures *Movement Disorders* **31** 1314–26
Kuhner A *et al* 2020 Motion biomarkers showing maximum contrast between healthy subjects and Parkinson's disease patients treated with deep brain stimulation of the subthalamic nucleus. a pilot study *Front. Neurosci.* **13** 1450
Kuruvithadam K *et al* 2021 Data-driven investigation of gait patterns in individuals affected by normal pressure hydrocephalus *Sensors* **21** 6451
Lam B *et al* 2021 Using wearable activity trackers to predict type 2 diabetes: –based cross-sectional study of the UK biobank accelerometer cohort *JMIR Diabetes* **6** e23364
Li N *et al* 2019 Monitoring motor symptoms in Parkinson's disease via instrumenting daily artifacts with inertia sensors *CF Trans. Pervasive Comput. Interaction* **1** 100–13
Lim A C Y *et al* 2022 The application of artificial intelligence and custom algorithms with inertial wearable devices for gait analysis and detection of gait-altering pathologies in adults: a scoping review of literature *Digit. Health* **8** 20552076221074128
Lindhiem O *et al* 2022 Objective measurement of hyperactivity using mobile sensing and machine learning: Pilot study *MIR Formative Res.* **6** e35803
Lonini L *et al* 2018 Wearable sensors for Parkinson's disease: which data are worth collecting for training symptom detection models *Npj Digit. Med.* **1** 64
Luštrek M *et al* 2021 A personal health system for self-management of congestive heart failure (HeartMan): development, technical evaluation, and proof-of-concept randomized controlled trial *JMIR Med. Inform.* **9** e24501
Ma C *et al* 2022a Quantitative assessment of essential tremor based on machine learning methods using wearable device *Biomed. Signal Process. Control* **71** 103244
Ma C *et al* 2022b Objective quantification of the severity of postural tremor based on kinematic parameters: a multi-sensory fusion study *Comput. Methods Programs Biomed.* **219** 106741
Madrid-Navarro C J *et al* 2018 Multidimensional circadian monitoring by wearable biosensors in Parkinson's disease *Front. Neurol.* **9** 157
Maleki G *et al* 2022 Objective monitoring of facioscapulohumeral dystrophy during clinical trials using a smartphone app and wearables: observational study *MIR Formative Res.* **6** e31775
Mannini A *et al* 2016 A machine learning framework for gait classification using inertial sensors: application to elderly, post-stroke and huntington's disease patients *Sensors* **16** 134
Mason A E *et al* 2022 Detection of COVID-19 using multimodal data from a wearable device: results from the first TemPredict study *Sci. Rep.* **12** 3463
McGinnis R S *et al* 2019 Rapid detection of internalizing diagnosis in young children enabled by wearable sensors and machine learning *PLOS ONE* **14** e0210267
Mehrang S *et al* 2020 Classification of atrial fibrillation and acute decompensated heart failure using smartphone mechanocardiography: a multilabel learning approach *IEEE Sensors J.* **20** 7957–68
Melin M *et al* 2016 Variability in physical activity assessed with accelerometer is an independent predictor of mortality in CHF patients *PLOS ONE* **11** e0153036
Memar S *et al* 2017 Segmentation and detection of physical activities during a sitting task in Parkinson's disease participants using multiple inertial sensors *J. Appl. Biomed.* **15** 282–90
Mesin L *et al* 2022 A multi-modal analysis of the freezing of gait phenomenon in Parkinson's disease *Sensors* **22** 2613
Mileti I *et al* 2018 Measuring gait quality in Parkinson's disease through real-time gait phase recognition *Sensors* **18** 919

Mirelman A *et al* 2021 Detecting sensitive mobility features for Parkinson's disease stages via machine learning *Movement Disorders* **36** 2144–55
Moon S *et al* 2020 Classification of Parkinson's disease and essential tremor based on balance and gait characteristics from wearable motion sensors via machine learning techniques: a data-driven approach *J. Neuroeng. Rehabil.* **17** 125
Morris S J and Paradiso J A 2003 A compact wearable sensor package for clinical gait monitoring, Offspring 1 7-15
Mostafa T A, Soltaninejad S, McIsaac T L and Cheng I 2021 A comparative study of time frequency representation techniques for freeze of gait detection and prediction *Sensors* **21** 6446
Moy M L *et al* 2014 Daily step count is associated with plasma c-reactive protein and IL-6 in a US cohort with COPD *Chest* **145** 542–50
Narziev N *et al* 2020 STDD: short-term depression detection with passive sensing *Sensors* **20** 1396
Ngo T *et al* 2021 Balance deficits due to cerebellar ataxia: a machine learning and cloud-based approach *IEEE Trans. Biomed. Eng.* **68** 1507–17
Noor M H M *et al* 2021 Detection of freezing of gait using unsupervised convolutional denoising autoencoder *IEEE Access* **9** 115700–9
Oliveira F H M, Machado A R P and Andrade A O 2018 On the use of it/i-distributed stochastic neighbor embedding for data visualization and classification of individuals with Parkinson's disease *Computat. Math. Methods Med.* **2018** 1–17
Oubre B *et al* 2021 Decomposition of reaching movements enables detection and measurement of ataxia *Cerebellum* **20** 811–22
Oung Q W *et al* 2017 Empirical wavelet transform based features for classification of Parkinson's disease severity *J. Med. Syst*. **42** 29
Palmerini L *et al* 2017 Identification of characteristic motor patterns preceding freezing of gait in Parkinson's disease using wearable sensors *Front. Neurol.* **8** 394
Papadopoulos A *et al* 2020 Unobtrusive detection of Parkinson's disease from multi-modal and in-the-wild sensor data using deep learning techniques *Sci. Rep.* **10** 21370
Papini G B *et al* 2020 Wearable monitoring of sleep-disordered breathing: estimation of the apnea–hypopnea index using wrist-worn reflective photoplethysmography *Sci. Rep.* **10** 13512
Pardoel S *et al* 2021 Early detection of freezing of gait during walking using inertial measurement unit and plantar pressure distribution data *Sensors* **21** 2246
Park B *et al* 2022 Modeling brain volume using deep learning-based physical activity features in patients with dementia *Front. Neuroinform.* **16** 795171
Park D J *et al* 2021 Evaluation for parkinsonian bradykinesia by deep learning modeling of kinematic parameters *J. Neural Transm.* **128** 181–9
Park E, Chang H J and Nam H S 2017 Use of machine leaning classifiers and sensor data to detect neurological deficit in stroke patients *J. Med. Internet Res.* **19** e120
Patel S *et al* 2009 Monitoring motor fluctuations in patients with Parkinson's disease using wearable sensors *IEEE Trans. Inf. Technol. Biomed.* **13** 864–73
Pedrelli P *et al* 2020 Monitoring changes in depression severity using wearable and mobile sensors *Fron. Psychiatry* **11** 584711
Ramesh V and Bilal E 2022 Detecting motor symptom fluctuations in Parkinson's disease with generative adversarial networks *Npj Digit. Med.* **5** 138
Rehman R Z U *et al* 2020 Accelerometry-based digital gait characteristics for classification of Parkinson's disease: what counts? *IEEE Open J. Eng. Med. Biol.* **1** 65–73
Rehman R Z U *et al* 2019 Comparison of walking protocols and gait assessment systems for machine learning-based classification of Parkinson's disease *Sensors* **19** 5363
Rehman R Z U *et al* 2022 Investigating the impact of environment and data aggregation by walking bout duration on Parkinson's disease classification using machine learning *Front. Aging Neurosci.* **14** 808518
Rodríguez-Martín D *et al* 2015 Posture transition identification on PD patients through a SVM-based technique and a single waist-worn accelerometer *Neurocomputing* **164** 144–53
Rodríguez-Martín D *et al* 2017 Home detection of freezing of gait using support vector machines through a single waist-worn triaxial accelerometer *PLOS ONE* **12** e0171764
Romijnders R *et al* 2022 A deep learning approach for gait event detection from a single shank-worn IMU: validation in healthy and neurological cohorts *Sensors* **22** 3859
Rovini E, Maremmani C and Cavallo F 2020 A wearable system to objectify assessment of motor tasks for supporting Parkinson's disease diagnosis *Sensors* **20** 2630
Rovini E *et al* 2018 Comparative motor pre-clinical assessment in Parkinson's disease using supervised machine learning approaches *Ann. Biomed. Eng.* **46** 2057–68
Sajal M S R *et al* 2020 Telemonitoring Parkinson's disease using machine learning by combining tremor and voice analysis *Brain Inform.* **7** 12
Samà A *et al* 2017 Estimating bradykinesia severity in Parkinson's disease by analysing gait through a waist-worn sensor *Comput. Biol. Med.* **84** 114–23
Samà A *et al* 2018 Determining the optimal features in freezing of gait detection through a single waist accelerometer in home environments *Pattern Recognit. Lett.* **105** 135–43
San-Segundo R *et al* 2020 Parkinson's disease tremor detection in the wild using wearable accelerometers *Sensors* **20** 5817
Scheid B H *et al* 2022 Predicting severity of huntington's disease with wearable sensors *Front. Digit. Health* **4** 874208
Shahzad A *et al* 2022 Automated prescreening of mild cognitive impairment using shank-mounted inertial sensors based gait biomarkers *IEEE Access* **10** 15835–44
Shawen N *et al* 2020 Role of data measurement characteristics in the accurate detection of Parkinson's disease symptoms using wearable sensors *J. Neuroeng. Rehabil.* **17** 52
Sigcha L *et al* 2020 Deep learning approaches for detecting freezing of gait in Parkinson's disease patients through on-body acceleration sensors *Sensors* **20** 1895
Sigcha L *et al* 2021 Automatic resting tremor assessment in Parkinson's disease using smartwatches and multitask convolutional neural networks *Sensors* **21** 291
Skibinska J and Burget R 2022 Is it possible to distinguish COVID-19 cases and influenza with wearable devices? Analysis with machine learning *J. Adv. Inf. Technol.* **13** 265–270
Sotirakis C *et al* 2022 Longitudinal monitoring of progressive supranuclear palsy using scpbody-worn/scp movement sensors *Movement Disorders* **37** 2263–71
Sun R *et al* 2021 Physical activity thresholds for predicting longitudinal gait decline in adults with knee osteoarthritis *Osteoarthritis Cartilage* **29** 965–72

Tabatabaei S A H *et al* 2020 Machine learning techniques for Parkinson's disease detection using wearables during a timed-up-and-go-test *Curr. Directions Biomed. Eng.* **6** 376–9
Talitckii A *et al* 2021 Avoiding misdiagnosis of Parkinson's disease with the use of wearable sensors and artificial intelligence *IEEE Sensors J.* **21** 3738–47
Talitckii A *et al* 2022 Comparative study of wearable sensors, video, and handwriting to detect Parkinson's disease *IEEE Trans. Instrum. Meas.* **71** 1–10
Teufl W *et al* 2021 Automated detection and explainability of pathological gait patterns using a one-class support vector machine trained on inertial measurement unit based gait data *Clin. Biomech.* **89** 105452
Thomas I *et al* 2018 A treatment-response index from wearable sensors for quantifying Parkinson's disease motor states *IEEE J. Biomed. Health Inform.* **22** 1341–9
Trabassi D *et al* 2022 Machine learning approach to support the detection of Parkinson's disease in IMU-based gait analysis *Sensors* **22** 3700
Vieira F G *et al* 2022 A machine-learning based objective measure for ALS disease severity *Npj Digit. Med.* **5** 45
Vijayan V *et al* 2021 Review of wearable devices and data collection considerations for connected health *Sensors* **21** 5589
Vos M D *et al* 2020 Discriminating progressive supranuclear palsy from Parkinson's disease using wearable technology and machine learning *Gait Posture* **77** 257–63
Wahle F *et al* 2016 Mobile sensing and support for people with depression: a pilot trial in the wild *MIR mHealth uHealth* **4** e111
Wan S *et al* 2018 Deep multi-layer perceptron classifier for behavior analysis to estimate Parkinson's disease severity using smartphones *IEEE Access* **6** 36825–33
Wang J *et al* 2022 Predicting adverse events during six-minute walk test using continuous physiological signals *Front. Physiol.* **13** 887954
Wasselius J *et al* 2021 Detection of unilateral arm paresis after stroke by wearable accelerometers and machine learning *Sensors* **21** 7784
Watts J *et al* 2021 Improving medication regimen recommendation for Parkinson's disease using sensor technology *Sensors* **21** 3553
West S L *et al* 2017 The association of daily activity levels and estimated kidney function in men and women with predialysis chronic kidney disease *Kidney Int. Rep.* **2** 874–80
Williamson J R *et al* 2021 Detecting Parkinson's disease from wrist-worn accelerometry in the U.K. Biobank *Sensors* **21** 2047
World Health Organization 1992 *The ICD-10 Classification of Mental and Behavioural Disorders: Clinical Descriptions and Diagnostic Guidelines* (World Health Organization)
Wu N *et al* 2021 Association between physical activity level and cardiovascular risk factors in adolescents living with type 1 diabetes mellitus: a cross-sectional study *Cardiovasc. Diabetol.* **20** 62
Xia Y *et al* 2018 A machine learning approach to detecting of freezing of gait in Parkinson's disease patients *J. Med. Imaging Health Inform.* **8** 647–54
Xing X *et al* 2022 Identification and classification of parkinsonian and essential tremors for diagnosis using machine learning algorithms *Front. Neurosci.* **16** 701632
Xu X *et al* 2021 Leveraging collaborative-filtering for personalized behavior modeling *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* vol 5, pp 1–27
Yin H *et al* 2021 DiabDeep: pervasive diabetes diagnosis based on wearable medical sensors and efficient neural networks *IEEE Trans. Emerg. Top. Comput.* **9** 1139–50
Zia S *et al* 2021 Detection of generalized tonic clonic seizures and falls in unconstraint environment using smartphone accelerometer *IEEE Access* **9** 39432–43